Useful XD Functions

User Guide

Access Limited Distribution

XD Insight

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Document Control

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1. Overview

1.1. Purpose of the Guide

The purpose of this document is to provide examples of useful XD functions. This includes:

- How to add functions to a report
- Function parameters
- Commonly used XD functions
- Manually editing functions

1.2. XD Functions Overview

XD functions are standard functions built into Insight. They perform operations using the provided parameters based on XD data fields (e.g. get email address, active pension scheme appointment data etc.). XD functions enable you to define logical columns returning the data without needing knowledge of the underlying data or internal logic.

Note:

- Some functions are only used for legacy report import purposes.
- To avoid complexity and improve performance, it is recommended to get data directly from data views where possible. Before using functions, check to see if the desired information can be returned through the standard views and columns.
- Ensure you use the Master table for the business area in your report. This will ensure you have maximum visibility of views and data for the business area.
- Some functions may not support multi-post sites, these are not included in this guide.

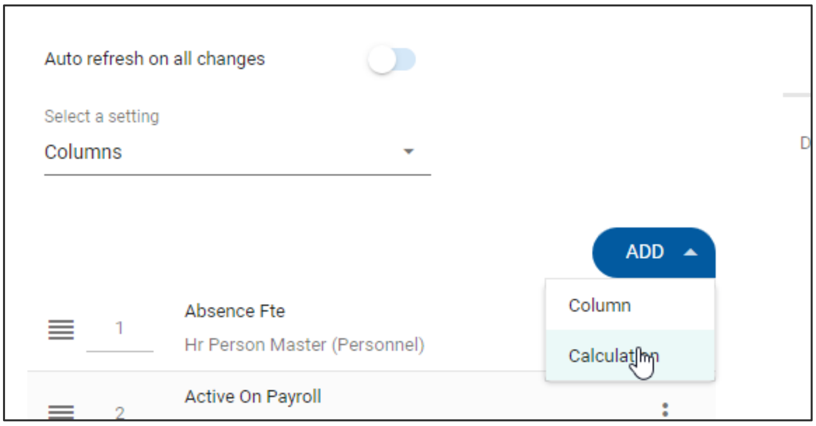
XD Functions are used in the following scenarios:

1. Abstract some of the logic for complex operations and data retrieval.
2. Getting data that may not accessible from the views in your report (for example, employee name or email address).

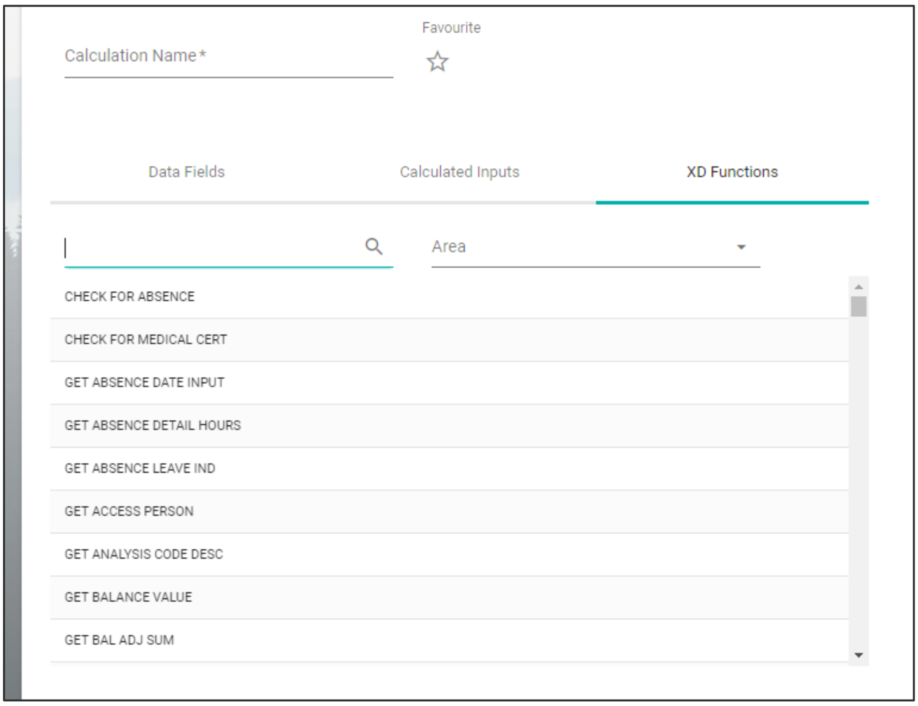
2. Adding Functions to a Report

To add an XD functions to a report, follow the below steps:

1. In Edit Mode, click **Add**.
2. Click **Calculation**.



3. Click the XD Functions tab. This will display a searchable list of functions. You can also filter by Business Area.



4. Click on the required function. A pop-up will then display the required parameters.
5. Populate the parameters. See section 2.1 for more detail.

Note: Mandatory parameters are marked with an asterisk.

6. Click **OK**.

2.1. Function Parameters

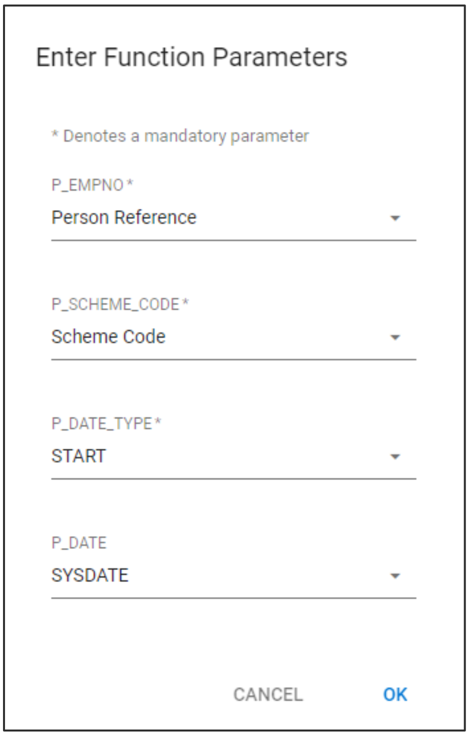
When adding a function, there are three ways to populate the Enter Function Parameters pop-up:

1. Take them from a column in your report (for example, person reference). Click the dropdown list for each parameter to see a list of the columns in your report, then click on the required column.
2. Populate with filter prompts in your report (for example, you might have a filter prompt for the dates for the scheme code).
3. Hard code the parameters. For example, if you want the date to be defined as the day of the report execution, you would use SYSDATE as the date parameter.

Note: It is recommended to review the function parameters required, and add the appropriate columns or filter prompts to the report before adding the function.

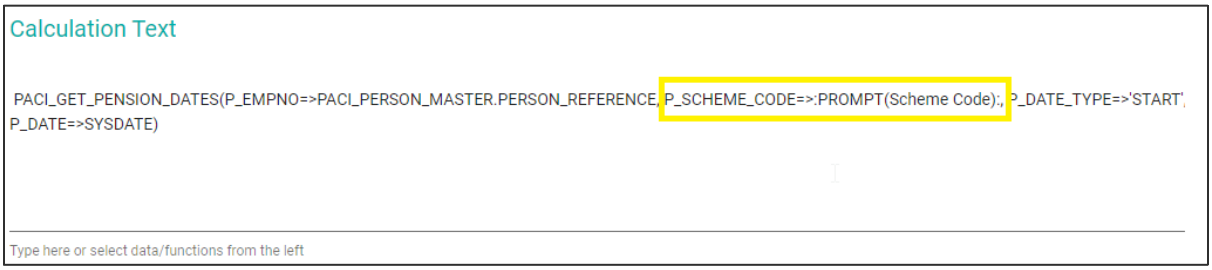
2.1.1. Function Parameters Example

The screenshot below displays an example of parameters entered for the *Get Pension Dates* function. This returns a date based on the person reference, scheme code, type of pension date you want (Start or End), and the Date you want to function to run against.

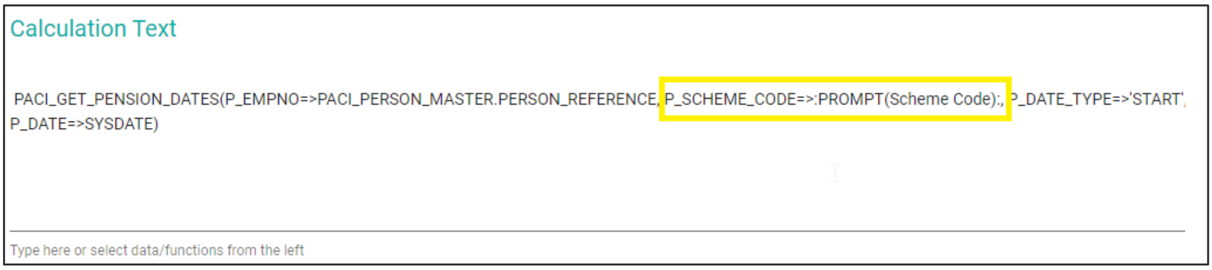


The parameters can be populated as follows:

- P_EMPNO – the Person Reference column from the report is selected from the dropdown list
- P_Scheme_Code - Scheme code is selected from the dropdown list.
  - When adding the function parameters, the system recognises that this column is associated with a filter prompt. Therefore, it configures the calculation to pass in the value from the prompt, as shown in the screenshot below:



  - In this example, if no filter prompt exists for *Scheme Code*, the system would take the value directly from the report column instead, as shown in the screenshot below:



- P_DATE_TYPE - START or END are manually typed in here.
- P_DATE – SYSDATE is manually typed in here. This is an Oracle SQL function that populates today's date (i.e. date of when report runs).

The above example will run the function taking:

1. The person reference from the report
2. Prompting for the Scheme Code, which has been set up as a filtered column
3. Hard coded date to *START*
4. Typed in *SYSDATE* command, which will pass in the current date at report run time.

The screenshow below displays the sample report highlighting the calculated column and output:


Auto refresh on all changes
Select a setting
Columns
ADD
1
Person Reference
Person Master (Pay)
2
active pension
Calculation
3
Scheme Code
Person Pension Scheme (Pay)
4
Pension Date
Calculation
Data as of 29-Mar-2023 16:06 (Record Count: 13)
Person Reference
Scheme Code
Pension Date
Scheme Code
2511250 16
2511167 16 01-Jun-2019
2511060 16 01-Apr-2019
2511170 16 01-Jun-2019
2511184 16 16-Jun-2019
2511216 16 01-Jun-2019
2511140 16 01-Jun-2019
2511153 16 16-Jun-2019
2511087 16 01-May-2019
2511090 16 01-May-2019
2511105 16 01-May-2019
2511119 16 01-May-2019
2511247 16 03-Jun-2019

3. General Guidelines

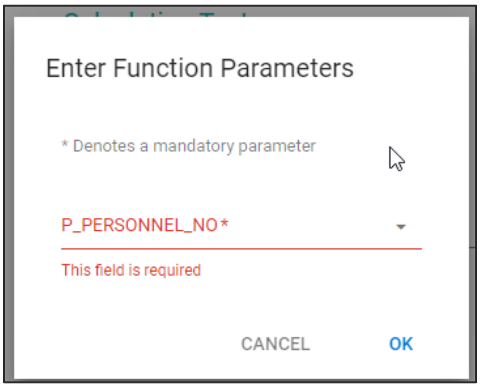
3.1. Security

Some XD functions require security access, depending on the area (e.g. HR system access, pay user, user defined fields access etc.). If you are missing the required permissions, the column may display without data.

For example, if a HR report contains a payroll function in a calculated column, and you only have access to 50% of the employees in payroll, that column will be blank for those employees who you do not have access to, but will be populated for the other 50%.

3.2. Parameters

All functions require you to populate some parameters. Some may be straight-forward, for example, a personnel reference such as *Get Full Name*. This can be populated from the *Person Reference* column in your report.



3.2.1. Date Parameters

Where you are asked for a date parameter, the standard format is DD-Mon-YYYY (e.g. 01-Jan-2023). The *SYSDATE* function can be used to populate a date parameter if you want to provide the date at report execution. See section 2.1.1 for an example of this.

4. Common XD Functions

The table on the following pages provides a list of useful XD functions, including a description of the parameter and what it is useful for.

Familiarity with your organisation's specific implementations of reference items (e.g. org reference codes, contact types, UDF codes etc.) is required to use some of these function. You may need to check XD reference screens (e.g. contact types. cost centres, department etc.) if unsure if what to populate.

Function name	Business area	Description	List of items
GET CONTACT TEXT	People Management	Get employee contact information for a specific contact type (e.g email, phone etc). **Parameters:** p empno: Employee number p contact type: Type of contact (MOBILE, EMAIL, HTN...) **Output:** Contact text	

Function name	Business area	Description	List of items
GET USER DATA FIELD	People Management	Get user code data (UDF) textfields. The user running this function requires personnel access to these fields. **Parameters:** p empno: Employee number p reference code: User Field Code p text: Text Column to return (1..12) p date: Date Column to return (1..2) **Output:** User data field value.	
CTBI GET PAY CODE HOURS	WFM	Return calculated hours based on dates and pay codes supplied. **Parameters:** p person: Employee number p from date: From Date p to date: To Date p pay code1: Pay code	

Function name	Business area	Description	List of items
		… p pay code15: Pay code **Output:** Return calculated hours for a list of pay codes (1..15) in a date period.	

Function name	Business area	Description	List of items
GET PRIM APPOINT DETAIL	People Management	Get primary appointment details for an employee on a particular day. **Parameters:** p personnel no: Employee number p date: Date p item: Data attribute to return (POST_SEQ, START_DATE, END_DATE...) **Output:** Return value of Item passed in as parameter.	POST_SEQ TO CHAR(post sequence) START_DATE TO CHAR(actual start date,DD-MON-RRRR) END_DATE to char(nvl(actual end date,ended date),DD-MON-RRRR) STATUS appointment status POST_TYPE actual post type POST_NO post number REASON reason code PEN_IND pensionable ind SEC secondment APT_ID appointment id DEPT department TARGET to char(target end date,DD-MON-RRRR) JOB_CATEGORY job category DIVISION division JOB_TITLE job title FTE_HOURS fte hours EMPLOYEE_STATUS employee status EMPLOYEE SUB STATUS employee sub status USER_CODE1 user code1 USER_CODE2 user code2

Function name	Business area	Description	List of items
			USER_CODE3 user code3 USER_CODE4 user code4 USER CODE5 user code5 MANAGEMENT_UNIT management unit COMPANY company COST CENTRE cost centre PROJECT project code CATEGORY category SUB_CATEGORY sub category HOURS hours WEEKS weeks FTE fte OVERRIDE FTE override fte ACTION action REPLACES_EMPLOYEE replacing employee COMMENTS comments PLANNED_END_DATE planned end date JOB_TITLE job title ABSENC_ FTE absence fte WOR_ GROUP work group

Function name	Business area	Description	List of items
GET APPOINTMENT DATA	People Management	Get appointment details for an employee on a particular day. **Parameters:** p appointment_id: Appointment ID p date: Date p item: Data attribute to display (POST_SEQ, START_DATE, END_DATE…) **Output:** Return value of Item passed in as parameter.	'APT_TYPE', 'POST_SEQ', 'START_DATE', 'END_DATE', 'STATUS', 'POST_TYPE', 'POST_NO', 'REASON', 'PEN_IND', 'SEC', 'DEPT', 'FTE', 'COMPANY', 'DIVISION', 'MGMT_UNIT', 'COST_CENTRE', 'LOCATION', 'WORK_GROUP', 'USER_FIELD1', 'USER_FIELD2', 'USER_FIELD3', 'USER_FIELD4', 'USER_FIELD5', 'JOB_TITLE', 'CATEGORY', 'EMPLOYEE_STATUS', 'SUB_STATUS', 'HOURS', 'FTE_HOURS',

Function name	Business area	Description	List of items
PABI GET MUL CODE PERIOD VALUE	Payroll	Get monetary value of one or more pay codes on a particular period. **Parameters:** p empno: Employee number p from period: Period from p to period: Period to p pay code1: Pay code ... p pay code15: Pay code **Output:** Return total monetary amount of the pay codes passed as parameters.	

Function name	Business area	Description	List of items
GET ADDRESS DETAIL	People Management	Get address details. **Parameters:** p person: Employee number p address type: Address type p detail: Detail of address to extract: *LINE1* *LINE2* *LINE3* *LINE4* *LINE5* *COUNTY* *COUNTRY* *POST CODE PRE* *POST CODE SUF* p ref code: Ref code for address type **Output:** Return the address detail of the p detail selected for an employee.	

Function name	Business area	Description	List of items
GET FULL NAME	People Management	Get employee's forename + surname. Useful for reporting from views that don't have employee name on them. **Parameters:** p personnel no: Employee number **Output:** Return full name of an employee.	
GET HR REFERENCE1 DESC	People Management	Get reference code description from reference1 codes. **Parameters:** p reference code: Reference code p reference type: Reference type code **Output:** Return description of reference code (e.g. company name)	Example of HR Reference 1 Items include company code (orgs), expense codes, locations. Generate a quick report of Insight view HR Reference1 (Reference Code, Reference Type), if you want to get a list of all these for reference.

Function name	Business area	Description	List of items
GET HR REFERENCE2 DESC	People Management	Get reference code description from reference2 codes. **Parameters:** p reference code: Reference code p reference type: Reference type code **Output:** Return description of reference code.	Example of HR Reference Items include Job Category (JOBCAT), Post Titles (POSTTL) Generate a quick report of Insight view HR Reference2 (Reference Code, Reference Type) if you want to get a list of all these for reference.

Function name	Business area	Description	List of items
RETRIEVE HR SALARY	People Management	Get values from different pay attributes based on date. **Parameters:** p empno: Employee number p date: Date p text: Attribute to display *PAYSCALE* *POINT* *MULTIPLIER* *AMOUNT* *RATETYPE* *PERSRATETYPE* *COMMENTS* **Output:** Return value of the attribute passed as parameter.	

Function name	Business area	Description	List of items
RETRIEVE APPOINT SALARY	People Management	Get salary attributes for an appointment. **Parameters:** p empno: Employee number p date: Date p text: Salary attributes *PAYSCALE* *POINT* *MULTIPLIER* *AMOUNT* *RATETYPE* *PERSRATETYPE* *COMMENTS* *DATE EFFECTIVE* p appointment id: Appointment id **Output:** Return value of the attribute passed as parameter.	

Function name	Business area	Description	List of items
GET CURRENT BALANCE	WFM	Get personal balance. **Parameters:** p person: Employee number p balance code: Balance code **Output**: Return current balance.	
GET PENSION DATES	Payroll	Get join/end date of a pension scheme. **Parameters:** p empno: Employee number p scheme code: Pension Scheme code p date type: START or END p date: Date **Output:** Return date of joining or end date depending on the date type passed as parameter.	

Function name	Business area	Description	List of items
RETRIEVE ROSTERED HOURS APPT ID	WFM	Get the number of rostered hours between two dates **Parameters:** p_person: Employee Number p_start_date: Date p_end_date: Date p_company_code: Company Code **Output:** Returns hours	

Function name	Business area	Description	List of items
RETRIEVE WORK PATTERN	WFM	Get the work pattern description associated with the employee **Parameters:** p_person: Employee Number p_start_date: Date p_end_date: Date p_company_code: Company Code (Number) p_appointment_id: appointmentID **Output** Work pattern Description	

Function name	Business area	Description	List of items
GET WORKED HOURS	Pay	Gets the hours from appointment based on the start date entered **Parameters** p_empno: Employee number p_date: Date **Output** Returns hours worked	

5. Manually Editing Functions

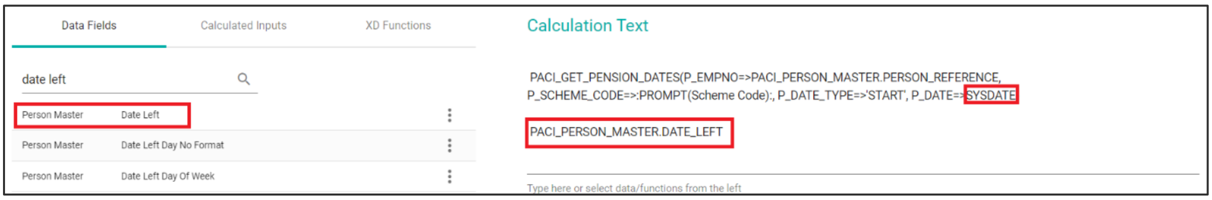
For more advanced use, once familiar with the syntax, you can make the following edits manually:

- change the columns
- change the parameters
- amend the filter prompts
- use another existing calculation as a parameter

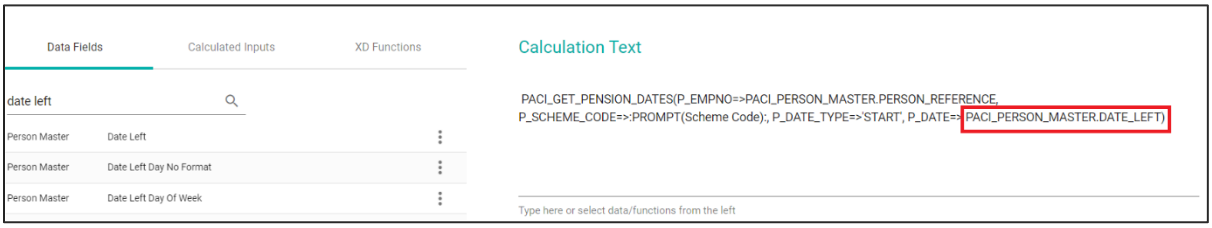
5.1. Changing parameters to other columns

The *Data Fields* tab displays all the other data in your report scope. In this example, if you wanted to change P_DATE param from *SYSDATE* to *date left* you would:

1. Click the required date column on the left. This is added to the end of the *Calculation Text* on the right.

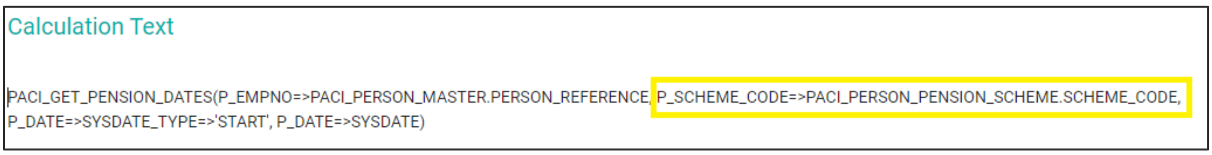


2. Delete *SYSDATE*
3. Insert the new *Date Left* field in its place.



5.2. Adding Prompts

The screenshot below displays an example of taking the scheme code directly from the column on the report.



This can be manually amended to work off the scheme code prompt.

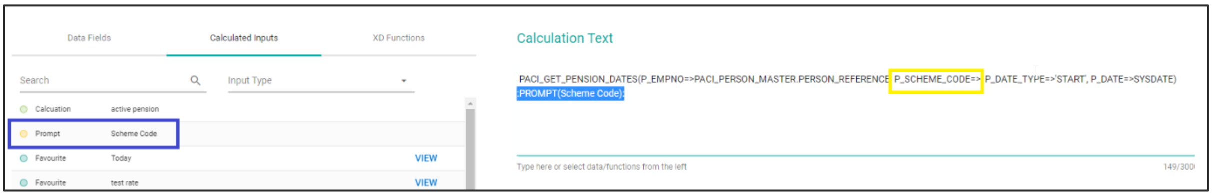
> **Note:** The prompt must already be set up in the filter settings.

The format is *:PROMPT(PROMPTNAME):*

In our example, *:PROMPT:(Scheme Code):* can be entered directly into the *P_Scheme_Code* field above, as follows:

PACI_GET_PENSION_DATES(P_EMPNO=>PACI_PERSON_MASTER.PERSON_REFERENCE, P_SCHEME_CODE=>:PROMPT(Scheme Code):, P_DATE_TYPE=>'START', P_DATE=>SYSDATE)

You can also amend it by clicking the prompt from the *Calculated Input* tab on the left, and then copying it into the P_SCHEME_CODE=> code variable.



5.3. Adding Calculations

If you have a calculated column producing a value that you which to be used as a parameter in your function, you can edit your current calculation so one of the parameters is fed by the other calculation.

The format is *:CALC(CALCNAME):*

You can amend this by clicking the existing calculation from the *Calculated Input* tab on the left, and then pasting it into the relevant parameter in your overall function.